Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 10.4 High-Frequency Response of Common-Gate and Cascode Amplifiers

Recap

$$F_H(s) = \frac{1 + a_1 s + a_2 s^2 + \cdots + a_n s^n}{1 + b_1 s + b_2 s^2 + \cdots + b_n s^n}$$

$$F_H(s) \simeq \frac{1}{1 + b_1 s}$$

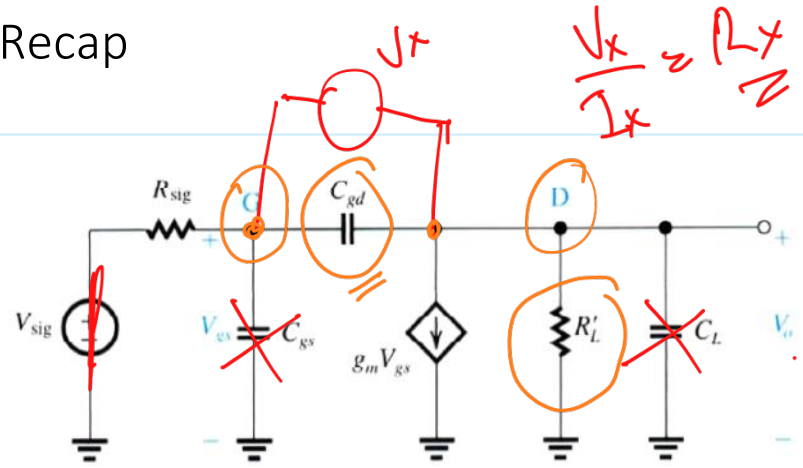
$$\omega_H = \frac{1}{b_1} \Rightarrow f_H = \frac{1}{b_1 \cdot 2\pi}$$

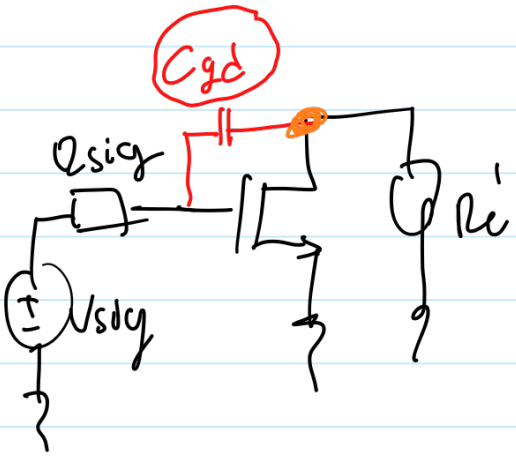
open-circuit time constants

$$b_1 = \sum_{i=1}^{n} C_i R_i \qquad b_1 = C_1 \cdot R_1 + C_2 R_2 + C_3 R_3$$

$$\omega_H = \frac{1}{b_1} = \frac{1}{\tau_H} = \frac{1}{\sum_i C_i R_i}$$

Recap



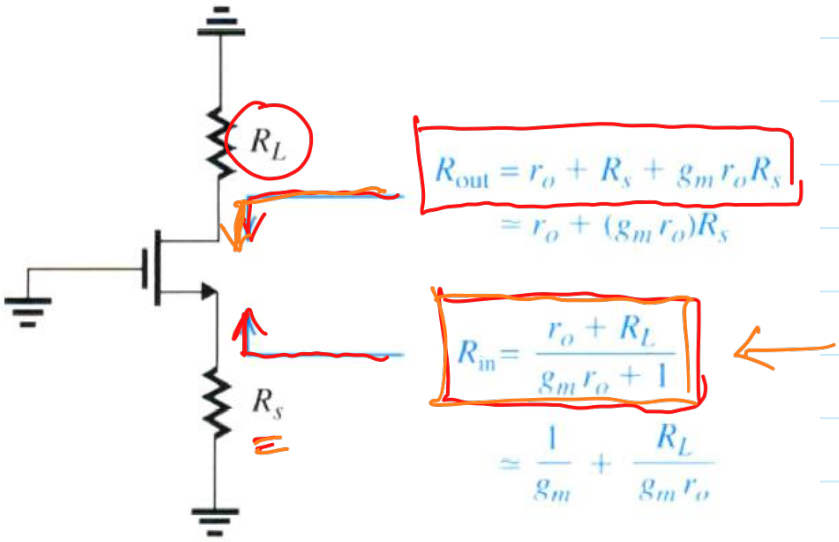


$$\tau_H = C_{gs}R_{gs} + C_{gd}R_{gd} + C_L R_{C_L}$$
$$= C_{gs}R_{sig} + C_{gd}[R_{sig}(1 + g_m R'_L) + R'_L] + C_L R'_L$$

$$R_{gd} = R_{sig}(1 + g_m R_L') + R_L'$$

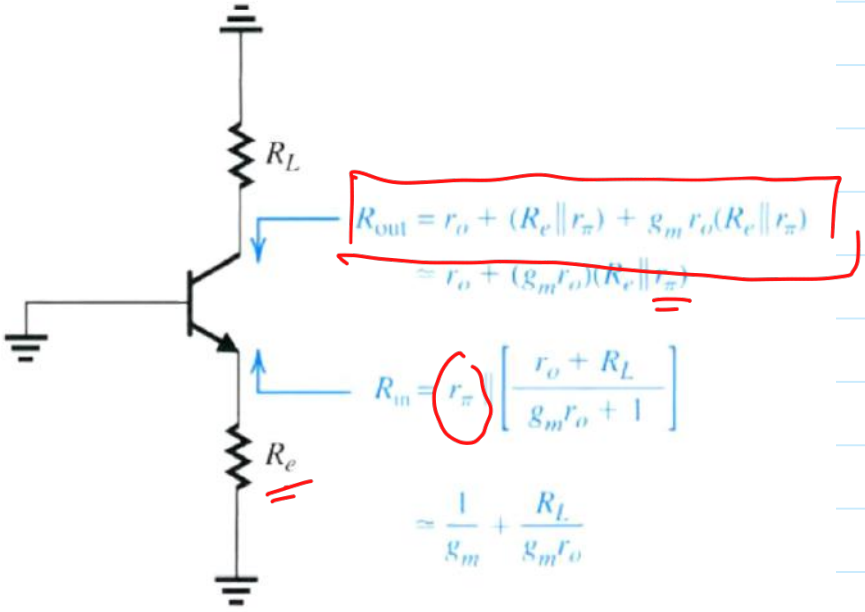
R_{gd} R_{gd}

Recap



$$R_{out} = r_o + R_s + g_m r_o R_s$$
$$= r_o + (g_m r_o) R_s$$

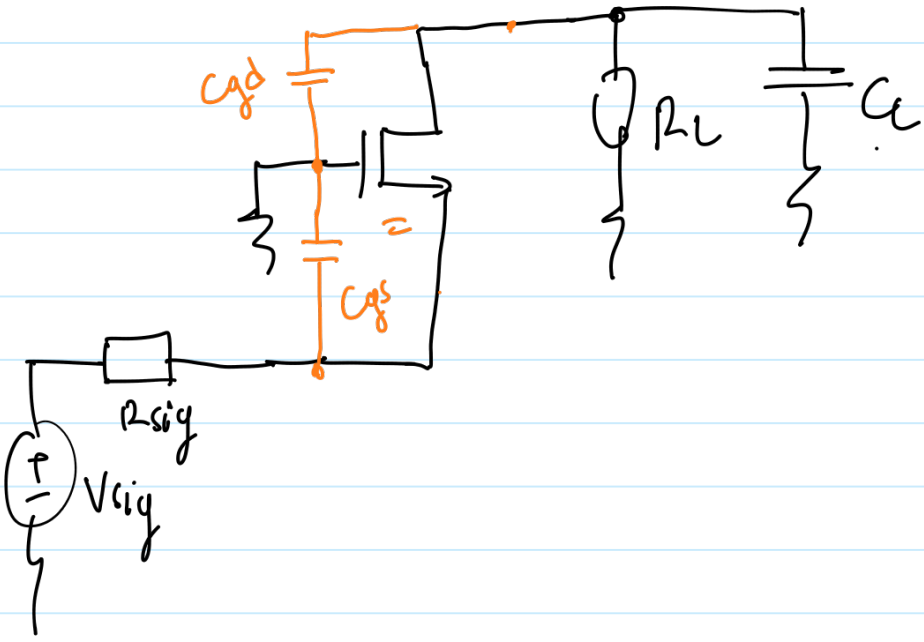
$$R_{in} = \frac{r_o + R_L}{g_m r_o + 1}$$
$$\simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

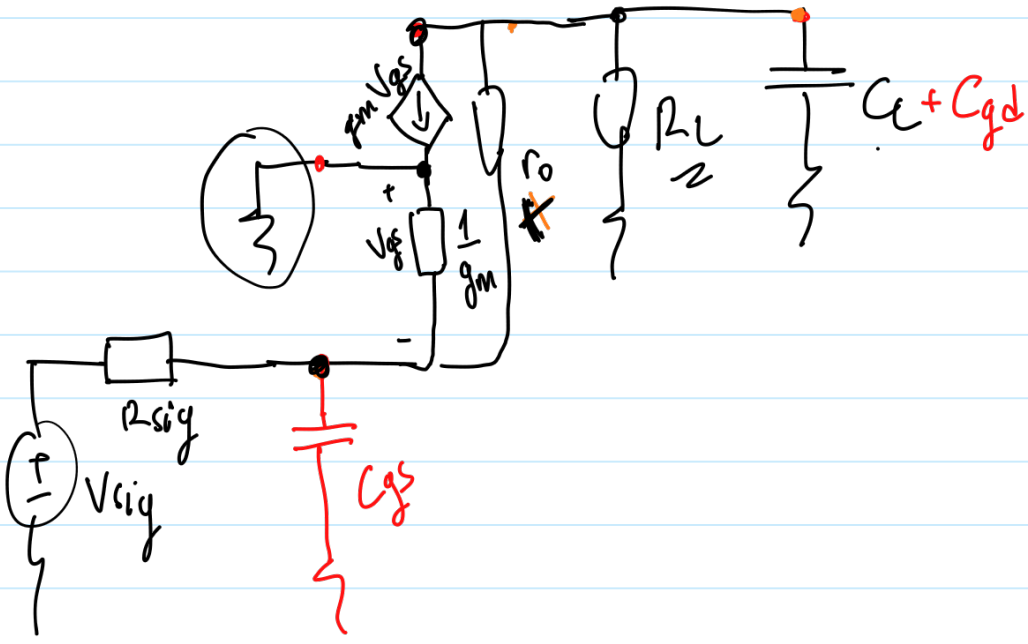


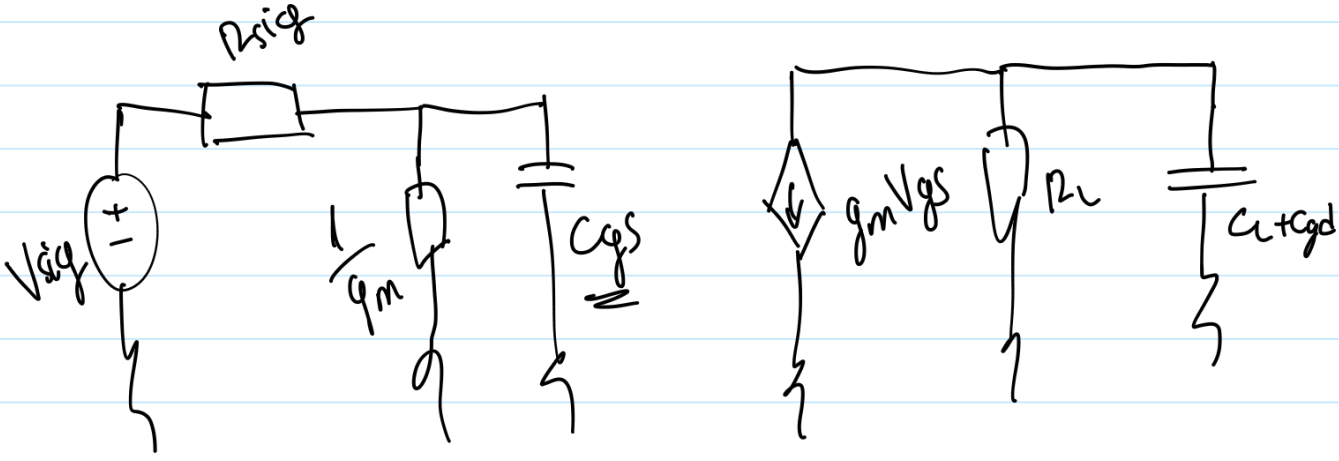
$$R_{out} = r_o + (R_e \| r_\pi) + g_m r_o (R_e \| r_\pi)$$
$$\simeq r_o + (g_m r_o)(R_e \| r_\pi)$$

$$R_{in} = r_\pi \| \left[\frac{r_o + R_L}{g_m r_o + 1} \right]$$
$$\simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

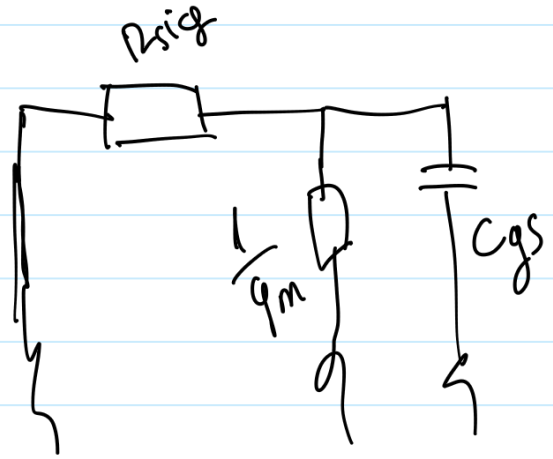
HF Response of Common Gate Amplifier







HF Response of Common Gate Amplifier



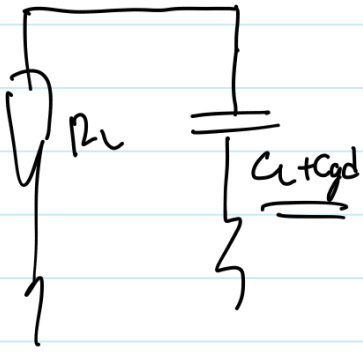
$$\Rightarrow R_{gs} = R_{sig} \,//\, 1/g_m$$

$$\tau_{H1} = R_{gs} \cdot C_{gs} = \left(R_{sig} \,//\, 1/g_m\right) C_{gs}$$

$$\omega_{p_1} = \frac{1}{\tau_1}$$

$$f_{p1} = \frac{1}{2\pi \cdot \left(R_{sig} \,//\, 1/g_m\right) C_{gs}}$$

$$\tau_{H2} = R_L \left(C_L + C_{gd}\right)$$

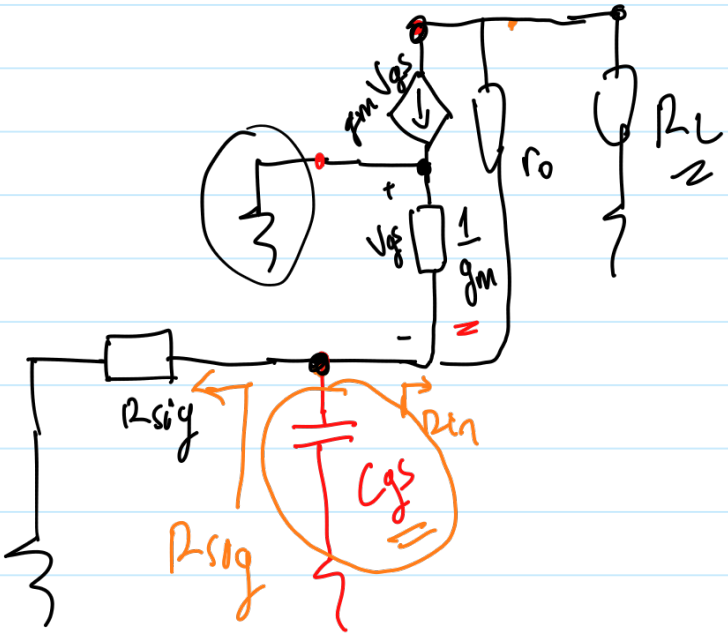


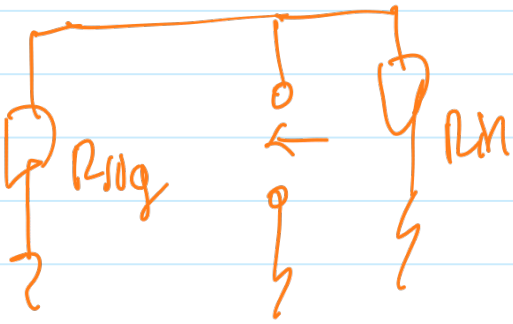
$$f_{p2} = \frac{1}{2\pi R_L \left(C_L + C_{gd}\right)}$$

$$\tau_H = \tau_{H1} + \tau_{H2}$$

$$b_1 = \frac{1}{\tau_H} = \omega_H$$

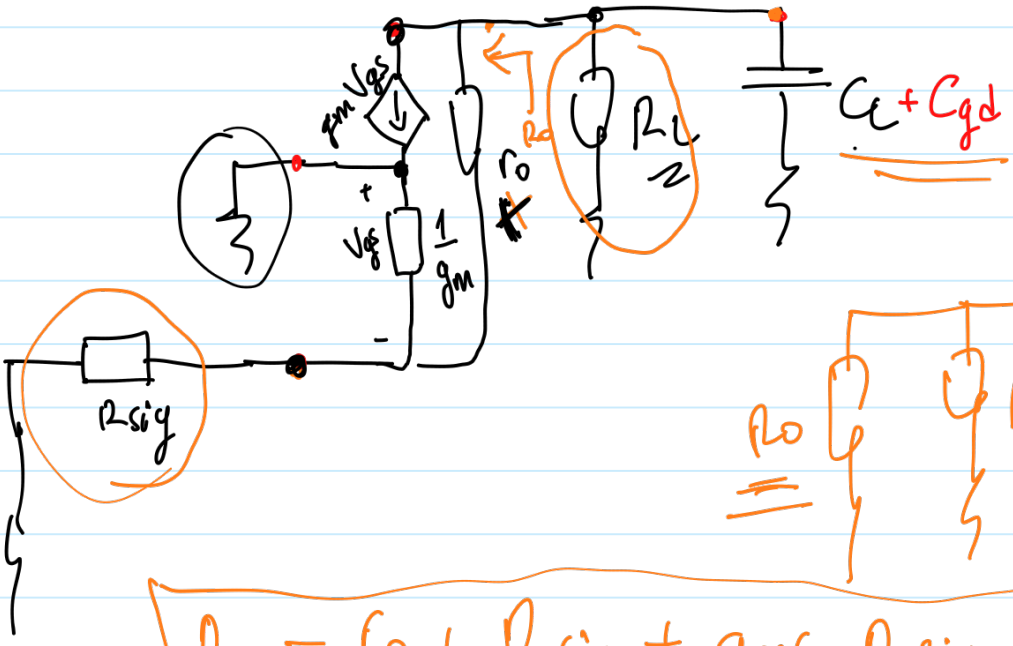
$$f_H = \frac{1}{2\pi \tau_H} = 1 \Big/ \left(\frac{1}{f_{p1}} + \frac{1}{f_{p2}}\right)$$

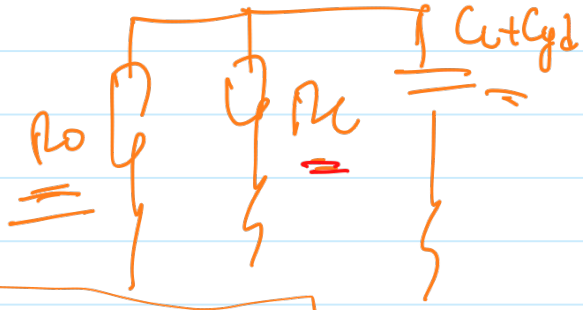




$$R_{in} = \frac{R_L + r_o}{1 + g_m r_o} \qquad R_{gs} = R_{sig} \,//\, R_{in}$$

$$\tau_{H1} = (R_{sig} \,//\, R_{in})\, C_{gs}$$

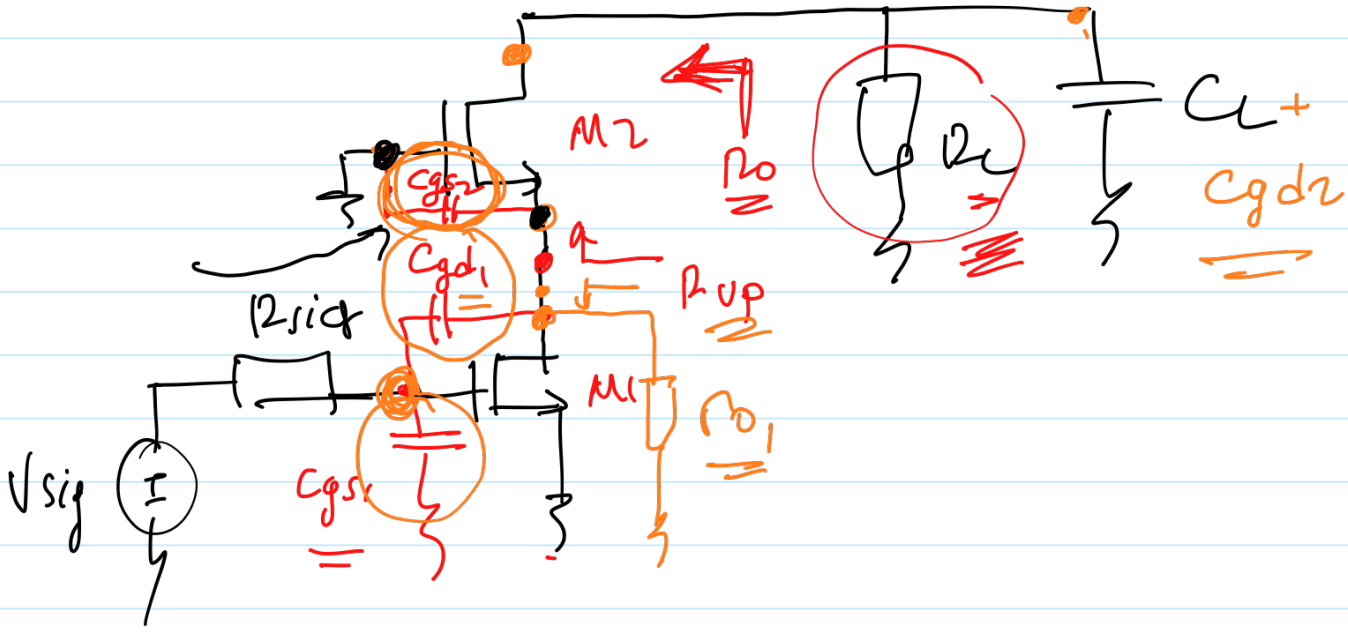




$$R_o = r_o + R_{sig} + g_m r_o R_{sig}$$

$$\tau_{H2} = (R_o \,//\, R_L)\,(C_c + C_{gd}) \qquad f_H = \frac{1}{2\pi \tau_H}$$

HF Response of Cascode Amplifier



$C_{gs1} \rightarrow R_{sig}$

$$R_{up} = \frac{r_{o2} + R_L}{g_{m2} r_{o2}}$$

$$R_{down} = r_{o1}$$

$$R_{d1} = r_{o1} \,//\, \left(\frac{r_{o2} + R_L}{g_{m2} r_{o2}} \right)$$

$$C_{gd1} \rightarrow R_{gd1} = \left(1 + g_{m1} R_{d1}\right) R_{sig} + R_{d1}$$

$$C_{gs2} + C_{db1} \rightarrow R_{d1}$$

$$C_L + C_{gd2} + C_{db2} \rightarrow R_L \,//\, \left(r_{o2} + r_{o1} + \left(g_m r_{o2}\right) r_{o1} \right)$$

HF Response of Cascode Amplifier

$$\tau_H = C_{gs_1} R_{sig} + C_{gd_1}\left[\left(1 + g_{m_1} R_{d_1}\right) R_{sig} + R_{d_1}\right] + R_{d_1}\left(C_{db_1} + C_{gs_2}\right) + \left(C_L + C_{gd_2} + C_{db_2}\right)\left(R_L /\!/ R_o\right)$$

$$f_H = \frac{1}{2\pi \cdot \tau_H}$$

Template